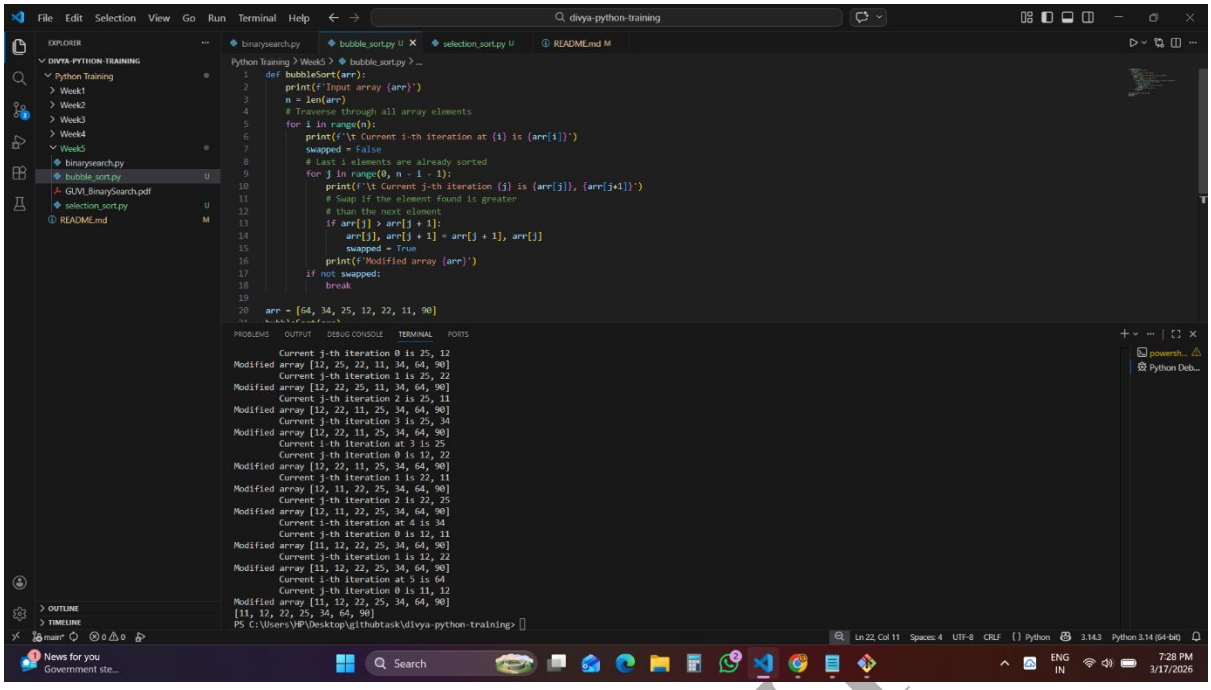
Bubble Sort:

```
def bubbleSort(arr):
    print(f'Input array {arr}')
    n = len(arr)
    # Traverse through all array elements
    for i in range(n):
        print(f'\t Current i-th iteration at {i} is {arr[i]}')
        swapped = False
        # Last i elements are already sorted
        for j in range(0, n - i - 1):
            print(f'\t Current j-th iteration {j} is {arr[j]}, {arr[j+1]}')
            # Swap if the element found is greater
            # than the next element
            if arr[j] > arr[j + 1]:
                arr[j], arr[j + 1] = arr[j + 1], arr[j]
                swapped = True
            print(f'Modified array {arr}')
        if not swapped:
            break

arr = [64, 34, 25, 12, 22, 11, 90]
bubbleSort(arr)
print(arr)
```

Divya.G (III-IT)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Week1
Week2
Week3
Week4
Week5
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bubble_sort.py
GUVI_BinarySearch.pdf
selection_sort.py
README.md
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bubble_sort.py U
selection_sort.py U
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Python Training > Week5 > bubble_sort.py > ...
def bubbleSort(arr):
print(f'Input array {arr}')
n = len(arr)
# Traverse through all array elements
for i in range(n):
print(f'\t Current i-th iteration at {i} is {arr[i]}')
swapped = False
# Last i elements are already sorted
for j in range(0, n - i - 1):
print(f'\t Current j-th iteration {j} is {arr[j]}, {arr[j+1]}')
# Swap if the element found is greater
# than the next element
if arr[j] > arr[j + 1]:
arr[j], arr[j + 1] = arr[j + 1], arr[j]
swapped = True
print(f'Modified array {arr}')
if not swapped:
break
arr = [64, 34, 25, 12, 22, 11, 90]
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Current j-th iteration 0 is 25, 12
Modified array [12, 25, 22, 11, 34, 64, 90]
Current j-th iteration 1 is 25, 22
Modified array [12, 22, 25, 11, 34, 64, 90]
Current j-th iteration 2 is 25, 11
Modified array [12, 22, 11, 25, 34, 64, 90]
Current j-th iteration 3 is 25, 34
Modified array [12, 22, 11, 25, 34, 64, 90]
Current i-th iteration at 3 is 25
Current j-th iteration 0 is 12, 22
Modified array [12, 22, 11, 25, 34, 64, 90]
Current j-th iteration 1 is 22, 11
Modified array [12, 11, 22, 25, 34, 64, 90]
Current j-th iteration 2 is 22, 25
Modified array [12, 11, 22, 25, 34, 64, 90]
Current i-th iteration at 4 is 34
Current j-th iteration 0 is 12, 11
Modified array [11, 12, 22, 25, 34, 64, 90]
Current j-th iteration 1 is 12, 22
Modified array [11, 12, 22, 25, 34, 64, 90]
Current i-th iteration at 5 is 64
Current j-th iteration 0 is 11, 12
Modified array [11, 12, 22, 25, 34, 64, 90]
[11, 12, 22, 25, 34, 64, 90]
PS C:\Users\HP\Desktop\githubtask\divya-python-training>
Ln 22, Col 11 Spaces: 4 UTF-8 CRLF {} Python 3.14.3 Python 3.14 (64-bit)
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